

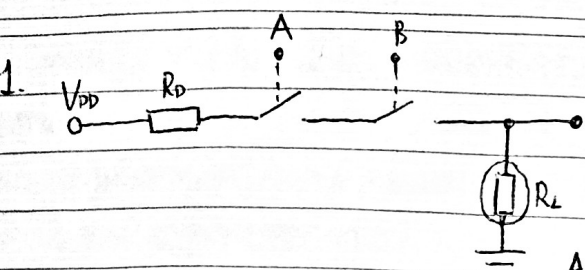
培道中學校

姓名： (號)

年級：

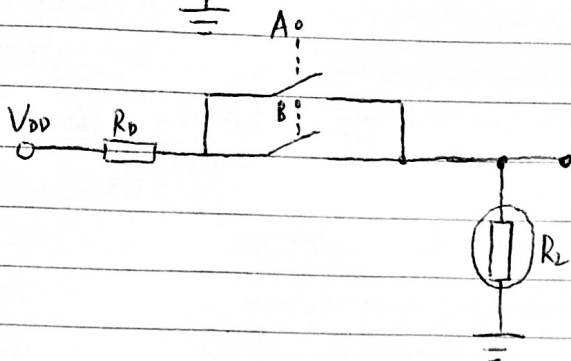
科目：

成績

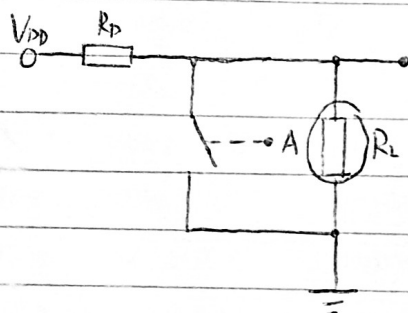


只有當 A、B 同時閉合時，燈泡才會亮
所以這是邏輯與功能

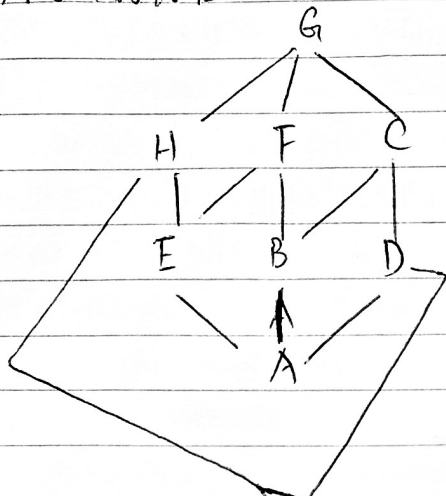
邏輯或功能



邏輯非功能



2. 找出三條由A到G的路徑



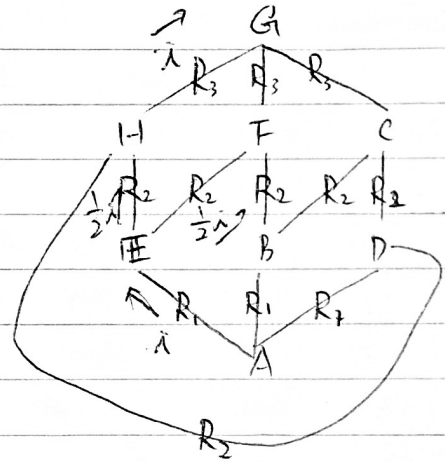
記 $V_G = 0$
 設 $V_H = V_E = V_D$, $A = \{H, F, C\}$
 $V_E = V_B = V_D$, $B = \{E, B, D\}$

$\forall i \in A, j \in B$

有 $R_{Ai} = \frac{(V_A - V_i)^2}{P}$, $R_{ij} = \frac{(V_i - V_j)^2}{P}$

$R_{jc} = \frac{V_j^2}{P}$

因此



$\lambda^2 R_3 = \lambda^2 R_1$

$\Leftrightarrow R_1 = R_3$

$(\frac{1}{2}\lambda)^2 R_2 = \lambda^2 R_1$

$\Leftrightarrow R_2 = 4R_1$

即有 $4R_1 = R_2 = 4R_3$

當 $V_A = 220V_{rms}$ 時, $V_A = \lambda R_1 + \frac{1}{2}\lambda R_2 + \lambda R_3 = 4\lambda R_1$

$P_A = 3\lambda^2 R_1 + 6(\frac{1}{2}\lambda)^2 R_2 + 3\lambda^2 R_3$

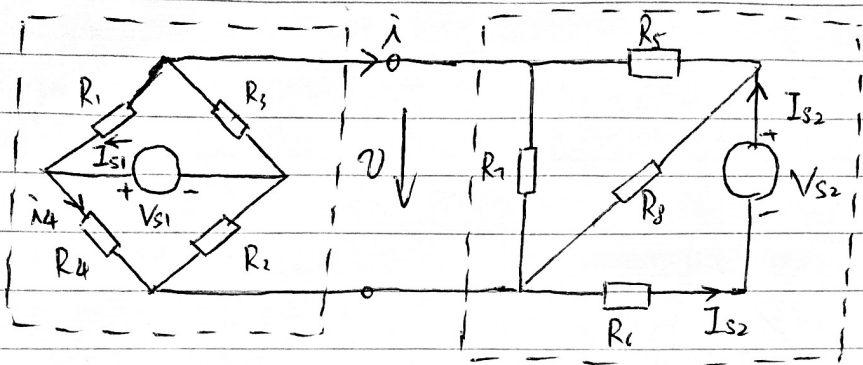
$= 6\lambda^2 R_1 + 6\lambda^2 R_1 = 12\lambda^2 R_1 = 3\lambda \cdot 4\lambda R_1 = \frac{3}{4} V_A^2 \cdot \frac{1}{R_1}$

$\Rightarrow R_1 = \frac{3}{4} \cdot \frac{V_A^2}{P_A} = \frac{3}{4} \cdot \frac{220^2}{1000} = 36.3 \Omega$

即 $R_{AB} = R_{AD} = R_{AE} = R_{HG} = R_{FG} = R_{CG} = 36.3 \Omega$

$R_{EH} = R_{BF} = R_{DC} = R_{EF} = R_{BC} = R_{DH} = 145.2 \Omega$

3.



原电路可看成两个有源网络对接

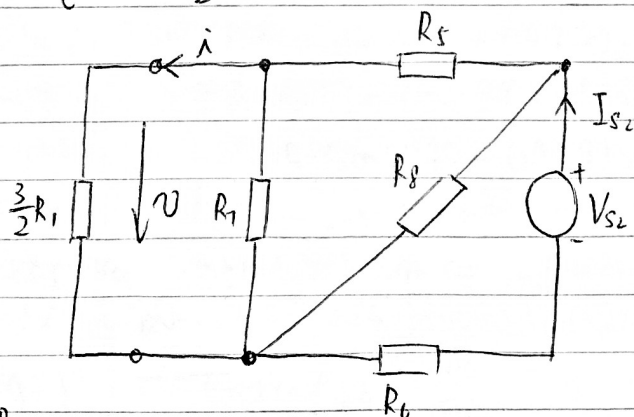
$$\Leftrightarrow \begin{cases} \lambda_4 R_4 + (\lambda + \lambda_4) R_2 = V_{S1} \\ (I_{S1} - \lambda_4) R_1 + (I_{S1} - \lambda_4 - \lambda) R_3 = V_{S1} \\ (I_{S1} - \lambda_4 - \lambda) R_1 - (I_{S1} - \lambda_4) R_3 \\ (I_{S1} - \lambda_4 - \lambda) R_3 - (\lambda + \lambda_4) R_2 = V \end{cases}$$

$$\Leftrightarrow \begin{cases} \lambda R_2 + \lambda_4 (R_2 + R_4) = V_{S1} \\ I_{S1} (R_1 + R_3) - \lambda R_3 - \lambda_4 (R_1 + R_3) = V_{S1} \\ I_{S1} R_3 - \lambda (R_2 + R_3) - \lambda_4 (R_2 + R_3) = V \end{cases} \quad (R_4 = R_2 = 2R_3 = 2R_1)$$

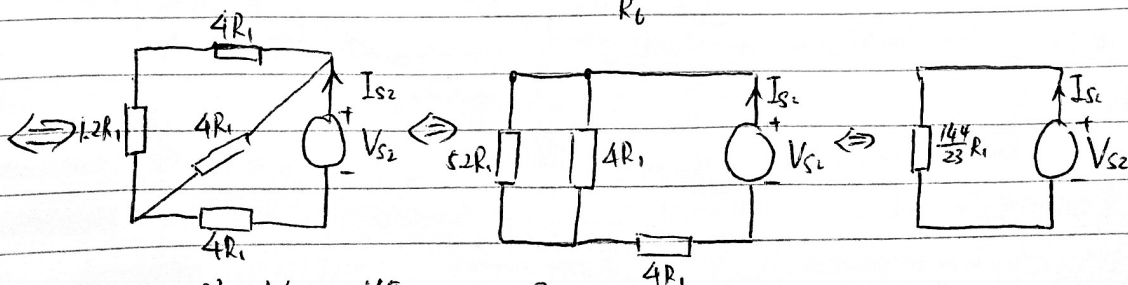
$$\Leftrightarrow \begin{cases} 2\lambda R_1 + 4\lambda_4 R_1 = V_{S1} \\ 2I_{S1} R_1 - \lambda R_1 - 2\lambda_4 R_1 = V_{S1} \\ I_{S1} R_1 - 3\lambda R_1 - 3\lambda_4 R_1 = V \end{cases} \Leftrightarrow \begin{cases} 4I_{S1} R_1 = 3V_{S1} \\ 4I_{S1} R_1 + 3\lambda R_1 = 3V_{S1} - 2V \end{cases}$$

$$\Leftrightarrow \begin{cases} I_{S1} = \frac{3}{4} \frac{V_{S1}}{R_1} \\ 3\lambda R_1 = -2V \end{cases} \Leftrightarrow \begin{cases} I_{S1} = \frac{3 \times 5V}{4 \times 1k\Omega} = 3.75mA \\ V = -\frac{2}{3} R_1 \cdot \lambda \end{cases}$$

因此原电路等效为



$$R_5 = R_8 = R_6 = 4R_1 \\ R_7 = 6R_1$$



因此, $I_{S2} = \frac{23}{144} \cdot \frac{V_{S2}}{R_1} = \frac{118}{144} \cdot \frac{5V}{1k\Omega} = \frac{23}{12} mA = 1.917mA$

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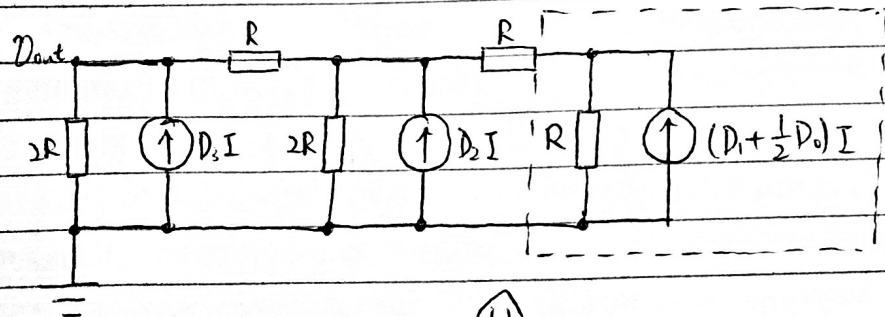
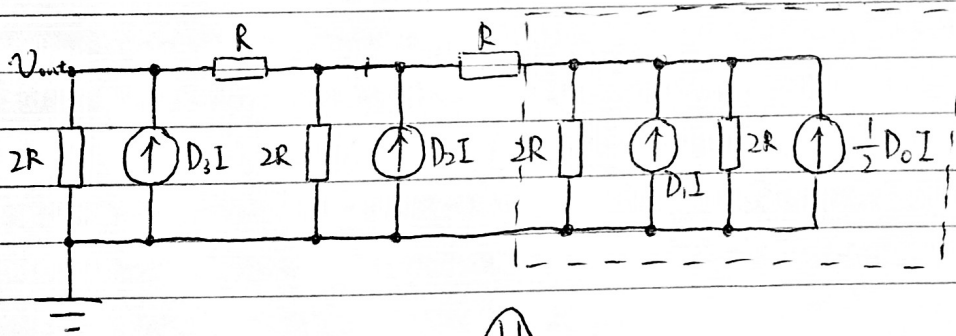
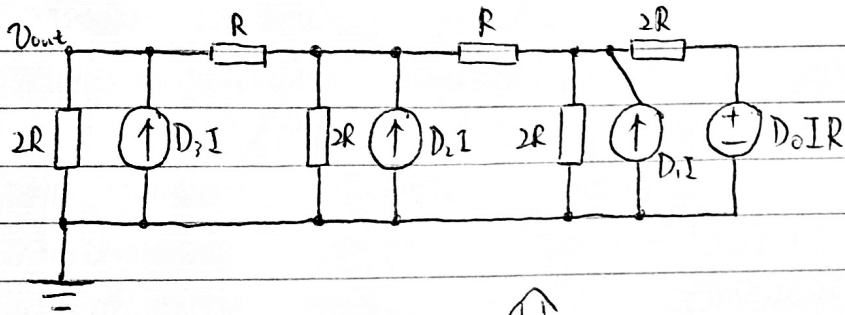
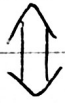
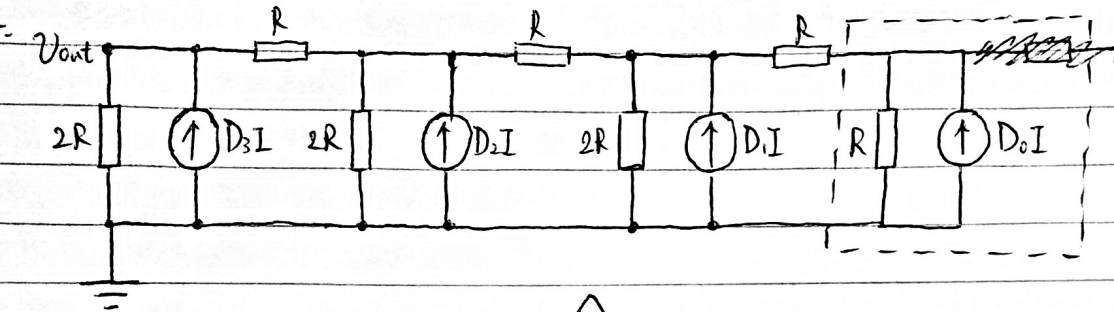
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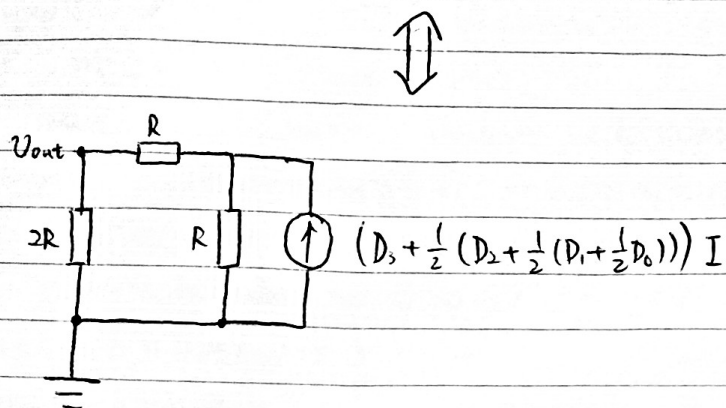
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因此, $V_{out} = \frac{1}{1+3} (D_3 + \frac{1}{2} (D_2 + \frac{1}{2} (D_1 + \frac{1}{2} D_0))) I \cdot 2R$

$$= \frac{IR}{2} (D_3 + \frac{1}{2} D_2 + \frac{1}{4} D_1 + \frac{1}{8} D_0)$$

$$= \frac{IR}{16} (2^3 D_3 + 2^2 D_2 + 2^1 D_1 + 2^0 D_0)$$

$$= \left(\sum_{n=0}^3 2^n D_n \right) \cdot \frac{IR}{16}$$

因此, 這是一個 4bit 的 DAC 轉換器

5. $\frac{V_1 - V_s}{R_s} + \frac{V_1 - V_2}{r_{be}} + \frac{V_1 - V_3}{R_p} = 0$

$$\frac{V_2 - V_1}{r_{be}} + \frac{V_2 - V_0}{R_E} + \frac{V_2 - V_3}{r_{ce}} + g_m (V_2 - V_1) = 0$$

$$\frac{V_3 - V_1}{R_p} + \frac{V_3 - V_2}{r_{ce}} + \frac{V_3 - V_0}{R_L} + g_m (V_1 - V_3) = 0 \quad (\text{假設 } V_0 = 0)$$

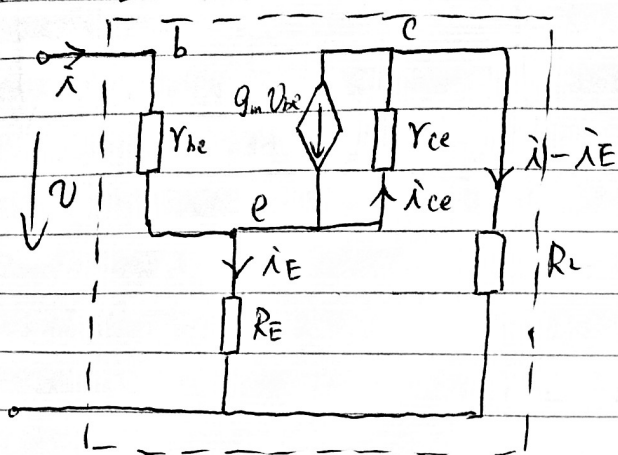
因此有

$$\begin{pmatrix} g_s + g_{be} + g_p & -g_{be} & -g_p \\ -(g_m + g_{be}) & g_m + g_{be} + g_E + g_{ce} & -g_{ce} \\ -g_p + g_m & -g_{ce} - g_m & g_p + g_{ce} + g_L \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \\ V_3 \end{pmatrix} = \begin{pmatrix} V_s g_s \\ 0 \\ 0 \end{pmatrix}$$

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6. 從輸入端口看入



$$v = i r_{be} + v_e$$

對 e 點列 KCL 方程: $i + g_m \cdot i r_{be} = i_e + i_{ce}$

$$\text{因此 } i_e R_E = [(1 + g_m r_{be}) i - i_e] r_{ce} + (i - i_e) R_L$$

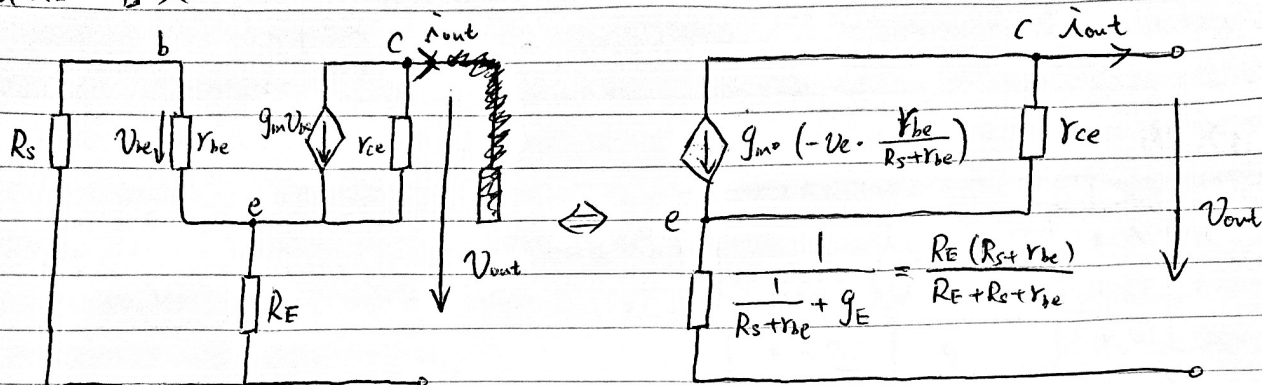
$$\Leftrightarrow i_e (R_E + r_{ce} + R_L) = (g_m r_{be} + 1) r_{ce} \cdot i + R_L \cdot i = [(g_m r_{be} + 1) r_{ce} + R_L] i$$

$$\text{因此 } v = i r_{be} + v_e = i r_{be} + i_e R_E = i r_{be} + \frac{R_E}{R_E + r_{ce} + R_L} \cdot i_e (R_E + r_{ce} + R_L)$$

$$= i r_{be} + \frac{R_E [(g_m r_{be} + 1) r_{ce} + R_L]}{R_E + r_{ce} + R_L} \cdot i$$

$$\text{即輸入電阻是 } R_{in} = r_{be} + \frac{R_E [(g_m r_{be} + 1) r_{ce} + R_L]}{R_E + r_{ce} + R_L}$$

從輸出端口看入



$$V_{out} = (-g_m - (-v_e \cdot \frac{r_{be}}{R_s + r_{be}}) - \lambda_{out}) r_{ce} + v_e$$

$$= (\frac{g_m r_{be} r_{ce}}{R_s + r_{be}} + 1) v_e - \lambda_{out} r_{ce}$$

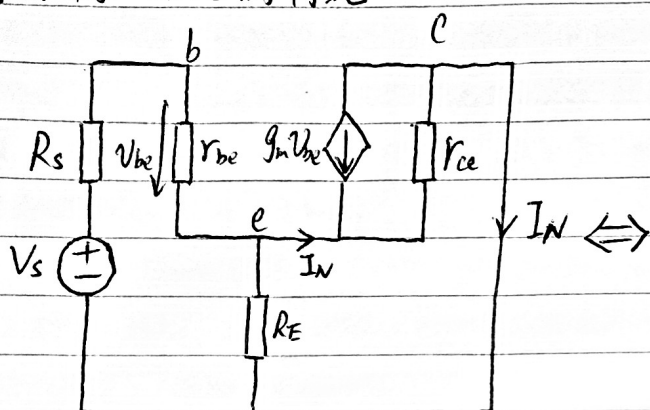
$$v_e = \left[g_m \cdot (-v_e \cdot \frac{r_{be}}{R_s + r_{be}}) + (-g_m - (-v_e \cdot \frac{r_{be}}{R_s + r_{be}}) - \lambda_{out}) \right] \frac{R_E (R_s + r_{be})}{R_E + R_s + r_{be}}$$

$$= -\lambda_{out} \cdot \frac{R_E (R_s + r_{be})}{R_E + R_s + r_{be}}$$

因此,
$$V_{out} = - \left(\frac{g_m r_{be} r_{ce}}{R_s + r_{be}} + 1 \right) \left(\frac{R_E (R_s + r_{be})}{R_E + R_s + r_{be}} \right) \lambda_{out} - r_{ce} \cdot \lambda_{out}$$

$$R_{out} = - \frac{V_{out}}{\lambda_{out}} = r_{ce} + \left(\frac{g_m r_{be} r_{ce}}{R_s + r_{be}} + 1 \right) \frac{R_E (R_s + r_{be})}{R_E + R_s + r_{be}}$$

考虑输出端短路的情况



$$V_s = \frac{V_{be}}{r_{be}} (R_s + r_{be}) - (I_N - \frac{V_{be}}{r_{be}}) R_E = \frac{V_{be}}{r_{be}} (R_s + r_{be} + R_E) - I_N R_E$$

$$\begin{aligned} 0 &= (I_N + g_m V_{be}) r_{ce} + (I_N - \frac{V_{be}}{r_{be}}) R_E \\ &= I_N (r_{ce} + R_E) + V_{be} (g_m r_{ce} - \frac{R_E}{r_{be}}) \\ &= I_N (r_{ce} + R_E) - \frac{V_{be}}{r_{be}} (R_E - g_m r_{be} r_{ce}) \end{aligned}$$

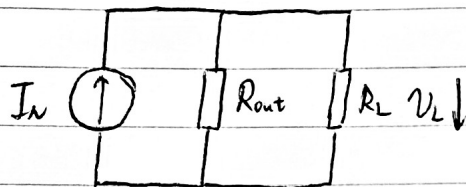
$$\Rightarrow \frac{V_{be}}{r_{be}} = \frac{I_N (r_{ce} + R_E)}{R_E - g_m r_{be} r_{ce}}$$

$$\Rightarrow V_s = \frac{I_N (r_{ce} + R_E)}{R_E - g_m r_{be} r_{ce}} (R_s + r_{be} + R_E) - I_N R_E$$

$$= I_N \left((R_s + r_{be} + R_E) \cdot \frac{r_{ce} + R_E}{R_E - g_m r_{be} r_{ce}} - R_E \right)$$

$$\text{因此, } I_N = \frac{V_s}{(R_s + R_E + r_{be}) \cdot \frac{r_{ce} + R_E}{R_E - g_m r_{be} r_{ce}} - R_E}$$

對於輸出端口，原網絡等效於

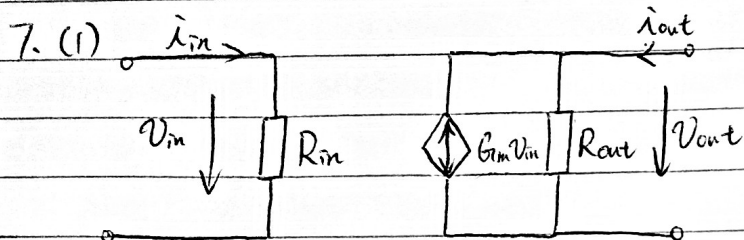


$$V_L = I_N \cdot \frac{R_L R_{out}}{R_L + R_{out}}$$

電壓放大倍數 $A_v = \frac{V_L}{V_s} = \frac{I_N}{V_s} \cdot \frac{R_L R_{out}}{R_L + R_{out}}$

$$= \frac{R_E - g_m r_{be} r_{ce}}{(R_s + R_E + r_{be})(r_{ce} + R_E) - R_E(R_E - g_m r_{be} r_{ce})} \cdot \frac{R_L R_{out}}{R_L + R_{out}}$$

$$A_v = \frac{R_E - g_m r_{be} r_{ce}}{(R_s + r_{be} + R_E)(r_{ce} + R_E) - R_E(R_E - g_m r_{be} r_{ce})} \cdot \frac{R_L \cdot \left[r_{ce} + \left(\frac{g_m r_{be} r_{ce}}{R_s + r_{be}} + 1 \right) \frac{R_E (R_s + r_{be})}{R_E + R_s + r_{be}} \right]}{R_L + r_{ce} + \left(\frac{g_m r_{be} r_{ce}}{R_s + r_{be}} + 1 \right) \frac{R_E (R_s + r_{be})}{R_E + R_s + r_{be}}}$$



$$\begin{aligned}
 P_{\Sigma} &= i_{in} V_{in} + i_{out} V_{out} \\
 &= V_{in}^2 G_{in} + (V_{out} G_{out} - G_m V_{in}) V_{out} \\
 &= G_{in} V_{in}^2 - G_m V_{in} V_{out} + G_{out} V_{out}^2
 \end{aligned}$$

(i) 當 $G_{in} < 0$ 或 $G_{out} < 0$ 時, 分別取 $V_{out} = 0$ 和 $V_{in} = 0$

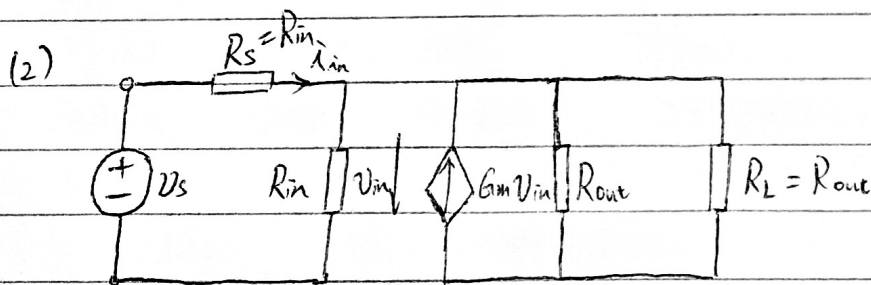
此時 $P_{\Sigma} = G_{in} V_{in}^2 < 0$, $P_{\Sigma} = G_{out} V_{out}^2 < 0$

(ii) 當 $G_{in} > 0$ 且 $G_{out} > 0$ 時

$$P_{\Sigma} = G_{in} \left(V_{in} - \frac{1}{2} \cdot \frac{G_m}{G_{in}} V_{out} \right)^2 + G_{out} V_{out}^2 - \frac{1}{4} \left(\frac{G_m}{G_{in}} \right)^2 V_{out}^2 \cdot G_{in}$$

$$= G_{in} \left(V_{in} - \frac{1}{2} \cdot \frac{G_m}{G_{in}} V_{out} \right)^2 + V_{out}^2 \left(G_{out} - \frac{1}{4} \cdot \frac{G_m^2}{G_{in}} \right)$$

當且僅當 $G_m^2 > 4 G_{in} G_{out}$ 時, 放大器有源。



$$G_{T,max} = \frac{P_{L,max}}{P_{S,max}} = \frac{\frac{1}{4} \cdot (G_m V_{in})^2 R_{out}}{\frac{1}{4} \cdot \frac{V_s^2}{R_{in}}} = \frac{G_m^2}{G_{out} G_{in}} \cdot \left(\frac{V_{in}}{V_s} \right)^2 = \frac{G_m^2}{4 G_{in} G_{out}}$$

當負載電阻 $R_L =$ 輸出電阻 R_{out} 時, 可向外輸出最大功率。